BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

BIOLOGY

Past Paper 2023 • Paper Code: Biology Paper–X (02) (23)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** Biology Paper–X (02) (23)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. An allele which completely marks the effect of other allele in a heterozygote:

(A) Dominant
(B) Recessive
(C) Co-dominant
(D) Incomplete-dominant

Q2. The process in which nitrates and nitrites are reduced to nitrogen gas by bacteria is.....

(A) Assimilation
(B) Nitrification
(C) De-nitrification
(D) Ammonifica-tion

Q3. Interferons are chemicals which are used to.....

(A) Kill virus
(B) Improve immunity
(C) Work as vaccine
(D) All these

Q4. Oral drugs are absorbed in the blood and are first taken to.....

(A) Liver
(B) Kidney
(C) Lungs
(D) Grain

Q5. Which one of the given controls the amount of light entering the eye?

(A) Pupil
(B) Iris
(C) Lense
(D) Ciliary body

Q6. Calcitonin hormone is secreted by.....

(A) Pituitary gland
(B) Thyroid gland
(C) Pancreas
(D) Gonads

Q7. The point where the muscle attaches to the moving bone is called.....

(A) Origin
(B) Insertion
(C) Extension
(D) Flexion

Q8. Which part of the flower contains the male reproductive structures called stamens?

(A) Calyx
(B) Corolla
(C) Androecium
(D) Gynoecium

Q9. Which respiratory disorder has permanent and irreversible effects?

(A) Bronchitis
(B) Emphysma
(C) Pneumonia
(D) Asthma

Q10. Which part of the air passage way is also called voice box?

(A) Bronchus
(B) Alveolus
(C) Larynx
(D) Trachea

Q11. On the concave side of the kidney there is a depression called.....

(A) The hilus
(B) The pelvis
(C) Ureter
(D) Renal capsule

Q12. Liver also plays a role in excretion. It makes urea from.....

(A) Uric acid
(B) Ammonia
(C) Carbondioxide
(D) Amino acids

Section-B

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 32

Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

QII. Answer any EIGHT parts. Each part carries FOUR marks.

1. Differentiate between the terms "respiration" and "breathing".
2. What are the different substances which plants excrete?
3. Define dialysis Relate the function of a dialyzer with that of the kidney.
4. Discuss the dual function of human ear.
5. Differentiate between nerve and neuron.

6. What is cartilage? Write different types of cartilage.

7. Write the adaptations of insect pollinated flowers.

8. Describe any four sources of variation.

9. What do you mean by conservation of nature? Write methods for the conservation of nature.

10. Write any four objectives of genetic engineering.

11. Differentiate between vaccines and antibiotics.

Section-C

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 21

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

Q. Attempt any THREE questions. All questions carry equal marks.

QIII.

(a) [4] What is nephron? Write the role of nephron in urine formation.

(b) [3] Draw label diagram of human respiratory system.

QIV.

(a) [4] What is chemical coordination? Discuss role of pancreas in chemical coordination.

(b) [3] Discuss three types of arthritis.

QV.

(a) [4] Define seed germination. Explain two types of seed germination.

(b) [3] Differentiate between chromosomes and chromatin.

QVI.

(a) [4] Define ecosystem. Discuss biotic components of ecosystem.

(b) [3] Write down how the following products are developed through fermentation: (i) Yogurt (ii) Bread (iii) Cheese